```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report
```

```
data=pd.read_csv("/content/IMDB Dataset.csv", engine='python')
print(data.head())
print(data.shape)
```

```
                                              review sentiment
0  One of the other reviewers has mentioned that ...  positive
1  A wonderful little production. <br /><br />The...  positive
2  I thought this was a wonderful way to spend ti...  positive
3  Basically there's a family where a little boy ...  negative
4  Petter Mattei's "Love in the Time of Money" is...  positive
(50000, 2)
```

```
X=data['review']
y=data['sentiment']

X_train, X_test, y_train, y_test=train_test_split(X, y, test_size=0.2, random_state=42)
```

```
vectorizer=CountVectorizer()
X_train_vec=vectorizer.fit_transform(X_train)
X_test_vec=vectorizer.transform(X_test)
```

```
model=MultinomialNB()
model.fit(X_train_vec, y_train)
```

```
▾ MultinomialNB
MultinomialNB()
```

```
y_pred=model.predict(X_test_vec)
```

```
print("Accuracy:",accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.8488
              precision    recall  f1-score   support

    negative       0.83      0.88      0.85      4961
    positive       0.87      0.82      0.85      5039

    accuracy                           0.85     10000
   macro avg       0.85      0.85      0.85     10000
weighted avg       0.85      0.85      0.85     10000
```

```
sample=["The movie was fantastic and exiciting"]
sample_vec=vectorizer.transform(sample)
prediction=model.predict(sample_vec)

print("Sentiment : ",prediction[0])
```

```
Sentiment :  positive
```

```
Start coding or generate with AI.
```